

Aditya Khadye

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Portfolio

PROFESSIONAL SUMMARY

Software Engineer building production AI and real-time, on-device systems in Python and Swift. Cut ETL runtime **39%** and drove **\$200K/yr in savings** at Travel + Leisure Co.; now building a real-time **visionOS** system with **on-device CoreML** at UCF's MIT² Lab and leading ML engineering for **ScriptGap**, a collaboration with Google, Adobe, and Monotype.

EDUCATION

University of Central Florida

Jan 2025 – Aug 2026

M.S. in Data Analytics (Machine Learning Track) · GPA 3.8 · Graduate Research Assistant (CECS/IE)

University of North Florida

Aug 2021 – Dec 2024

B.S. in Computer Science Honors · Dean's List x5 · IEEEExtreme 17.0 (47th in US, 2nd at UNF)

PROFESSIONAL EXPERIENCE

Graduate Research Assistant (Lead Researcher)

Jun 2025 – Present

University of Central Florida · TRC / MIT² Lab, Orlando, FL

- Engineered the **visionOS** client in Swift/SwiftUI for an **Apple Vision Pro** AR HUD, with an **on-device CoreML** inference path classifying driver-behavior and cognitive-load events locally; built a parallel cloud path to compare on-device versus server latency and privacy trade-offs.
- Built a low-latency **60 Hz** telemetry pipeline streaming vehicle data over UDP from a driving simulator, with a parallel TCP channel persisting time-locked session data, using binary packet parsing and actor-isolated concurrency to sustain real-time throughput on the headset.
- Lead data and ML engineering for **ScriptGap**, a Readability Consortium collaboration with Google, Adobe, and Monotype: built Transformer-based embedding pipelines and applied latent-space analysis over font (OTF) and web-reading (CrUX) data to rank global writing systems by typographic support, informing partner R&D.
- Building **Durandal**, a real-time data-quality agent monitoring multi-stream session data (telemetry, gaze, stimulus) across three machines, combining rule-based checks with Claude API reasoning and NTP/PTP cross-machine clock synchronization.
- Co-built the lab's research compute environment and deployed distributed workloads on Linux clusters (Slurm + MPI), reducing experiment reruns by **40%** via automated validation gates and runbooks.

Data Science Intern

May 2024 – Aug 2024

Travel + Leisure Co., Orlando, FL

- Led development of **DataPulse**, custom ETL/ELT workflows across the company's data lake (Snowflake, SQL Server, Informatica/IICS), cutting ETL runtime by **39%**.
- Built and deployed tree-based models (XGBoost/LightGBM) to production on Oracle with Redis for low-latency serving to forecast pipeline failures, reducing failed jobs by **28%**.
- Cut engineering manual workload by **15–25 hours/week (\$200K/yr in savings)** and improved data quality and system stability across Snowflake and Oracle environments.

KEY PROJECTS

Agentic AI Document Analysis Framework: mcp-clinical-doc-agent

May 2026

Python, FastAPI, LangGraph, Anthropic MCP, Claude Code, AWS, Pandas, Pydantic v2

- Built a FastAPI backend + **4-tool MCP server** consumed by a **Claude Code** agent for end-to-end document analysis, orchestrated via a **LangGraph** state machine, with a deterministic pass/fail eval gate (6 typed checks) to manage false-positive cost.
- Containerized on AWS (multi-stage Docker, healthchecks, ECR-ready) with Pandas-based ETL outputting join-ready CSVs; open-sourced on GitHub with CI, tests, and demo.

TrendScope: Market Analytics & LLM Narrative Engine

Mar 2026

Python, DuckDB, yfinance, Anthropic API, Pandas

- Built an idempotent ingestion pipeline pulling daily OHLCV for 15 equities and sector ETFs via yfinance into **DuckDB**, using composite-PK upserts and a per-run audit table for reproducible local analytics.
- Implemented a deterministic signal engine spanning trend, momentum, mean-reversion, and volatility indicators (RSI, ADX, multi-horizon momentum, realized-vol percentiles, relative strength vs. sector), with strict **point-in-time correctness** to prevent look-ahead bias.
- Integrated the **Anthropic API** with structured prompts combining signal output and ticker-level news to generate daily markdown digests, orchestrated end-to-end (ingest → signals → digest) via a single `make daily` command.

ViT (Vision Transformer): Training Performance Optimization

Apr 2025

PyTorch, Transformers, GPU Training, Mixed Precision, Input Pipelines

- Increased ViT training throughput to **1,240 images/sec** on a single RTX 5080 and raised GPU utilization to **92%** via mixed-precision training and input-pipeline tuning, cutting total training time by **36%**.

TECHNICAL SKILLS

Languages: Python, Swift/SwiftUI, C++, Java, SQL (PostgreSQL/Snowflake), R

Agentic AI: MCP (Model Context Protocol), Claude Code, LangGraph, FastAPI agent backends, RAG, tool orchestration, evaluation guardrails

AI/ML/NLP: Foundation models, LLMs, Transformers, NLP, on-device ML (CoreML), PyTorch, Hugging Face, embeddings/vector search, scikit-learn, XGBoost/LightGBM

MLOps/DevOps: CI/CD (Jenkins, GitHub Actions, ArgoCD), Terraform, Docker, Kubernetes, OpenShift, Slurm

Data Engineering: Snowflake (optimization/modeling), dbt, Airflow, ETL/ELT, real-time streaming, Azure, AWS (S3, EC2, IAM), Redis, Spark, BigQuery, NoSQL

Systems/Tools: Linux, Git, Swift concurrency (actors), visionOS/RealityKit, REST APIs, FastAPI, distributed computing (MPI), Agile/Scrum

CERTIFICATIONS

IBM – CI/CD (2026) · Microsoft – Azure SQL (2026) · Google – Agile & Scrum (2025) · IBM – ML with Python (2024) · Amazon – ML on AWS (2024) · Stanford

Online – Advanced Learning Algorithms (2024)